5.3 Principles and Components of Fourth-generation Drones

The principles and components of fourth-generation drones define the most advanced stage in unmanned aerial technology, where autonomy, intelligence, and connectivity converge to enable fully self-governing and adaptive aerial systems. Unlike previous generations that relied on remote human control, fourth-generation drones operate on the foundation of AI-driven autonomy, sensor fusion, and networked interoperability, allowing them to execute complex missions across air, land, sea, and space domains with minimal human intervention.

5.3.1 Principles of operation

Fourth-generation drones operate on four interdependent principles:

- *Perceive fuse decide act*: Continuous sensor-driven perception (multimodal sensing) is fused into a coherent situational picture; AI/ML-based decision layers convert high-level objectives into executable plans; and the flight/control stack executes and adapts those plans in real time.
- *Distributed, cooperative autonomy*: Intelligence is shared among platforms (swarm or team), enabling task allocation, formation control, cooperative sensing, and collaborative engagement without centralized micromanagement.
- *Edge-centric, low-latency processing*: Decision-quality processing is performed on or near the platform (edge computing) to permit immediate responses (collision avoidance, dynamic re-tasking) and to reduce bandwidth for transmitted data.
- *Resilience and graceful degradation*: Built-in redundancy, fault-detection-isolation-and-recovery (FDIR), anti-spoofing/jamming measures, and predefined safe-states allow continued operation or safe